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مدرسة الخالدية الحلقة الثالثة بنين الشرقية

AL KHALIDIA BOYS' SCHOOL · C3 EASTERN

الميثاق الوطني للتعليم

END OF TERM 1 · ACADEMIC YEAR 2026 – 2027

Grade 12 Advanced

Exam Coverage Compilation

تجميعة هيكل امتحان نهاية الفصل الدراسي الأول

Bridge to Calculus | 26 questions | MCQ 1–20 · FRQ 21–26 | 150 min · /100

Draft coverage — for teachers' feedback only · not for students

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بالميثاق نبلغ الآفاق

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◆ LESSON 2-1 · Tangent Lines and the Length of a Curve

PART 1 / 4

In exercises 7–12, estimate the length of the curve $y = f(x)$ on the given interval using (a) $n = 4$ and (b) $n = 8$ line segments. (c) If you can program a calculator or computer, use larger n's and conjecture the actual length of the curve.

7. $f(x) = \cos x, 0 \le x \le \pi/2$

(a) $n = 4$

(b) $n = 8$

8. $f(x) = \sin x, 0 \le x \le \pi/2$

(a) $n = 4$

(b) $n = 8$

9. $f(x) = \sqrt{x+1}, 0 \le x \le 3$

(a) $n = 4$

(b) $n = 8$

10. $f(x) = 1/x, 1 \le x \le 2$

(a) $n = 4$

(b) $n = 8$

11. $f(x) = x^2 + 1, -2 \le x \le 2$

(a) $n = 4$

(b) $n = 8$

12. $f(x) = x^3 + 2, -1 \le x \le 1$

(a) $n = 4$

(b) $n = 8$

◆ LESSON 2-3 · Computation of Limits

In exercises 1–28, evaluate the indicated limit, if it exists.

Assume that $\lim_{x\to 0} \frac{\sin x}{x} = 1$.

9. $\lim_{x\to 0} \frac{\sin x}{\tan x}$

10. $\lim_{x\to 0} \frac{\tan x}{x}$

11. $\lim_{x\to 0} \frac{xe^{-2x+1}}{x^2+x}$

THEOREM 3.5 (Squeeze Theorem)

Suppose that

$$f(x) \le g(x) \le h(x)$$

for all x in some interval (c, d), except possibly at the point $a \in (c, d)$ and that

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L,$$

for some number L. Then, it follows that

$$\lim_{x \to a} g(x) = L, \text{ also.}$$

LESSON 2-4 · Continuity and Its Consequences

THEOREM 4.4 (Intermediate Value Theorem)

Suppose that f is continuous on the closed interval $[a, b]$ and W is any number between $f(a)$ and $f(b)$. Then, there is a number $c \in [a, b]$ for which $f(c) = W$.

Theorem 4.4 says that if f is continuous on $[a, b]$, then f must take on *every* value between $f(a)$ and $f(b)$ at least once. That is, a continuous function cannot skip over any numbers between its values at the two endpoints. To do so, the graph would need to leap across the horizontal line $y = W$, something that continuous functions cannot do. (See Figure 2.29a.) Of course, a function may take on a given value W more than once. (See Figure 2.29b.) Although these graphs make this result seem reasonable, the proof is more complicated than you might imagine and we must refer you to an advanced calculus text.

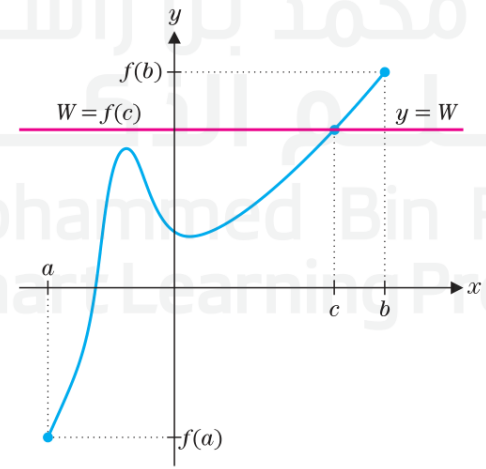


FIGURE 2.29a
An illustration of the Intermediate Value Theorem

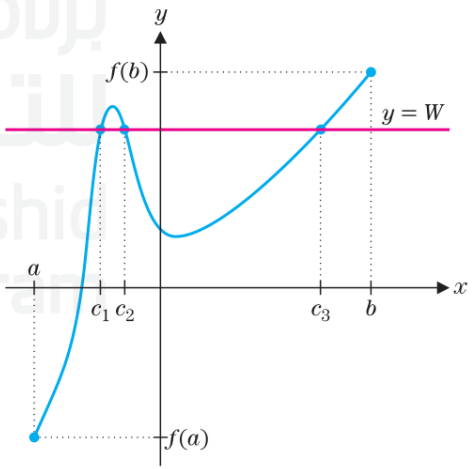


FIGURE 2.29b
More than one value of c

EXAMPLE 4.2 Removing a Hole in the Graph

Extend the function from example 4.1 to make it continuous everywhere by defining it at a single point.

Solution In example 4.1, we saw that the function is continuous for $x \neq 1$ and it is undefined at $x = 1$. So, suppose we just go ahead and define it, as follows. Let

$$g(x) = \begin{cases} \dfrac{x^2 + 2x - 3}{x - 1}, & \text{if } x \neq 1 \\ a, & \text{if } x = 1, \end{cases}$$

for some real number a.

Notice that $g(x)$ is defined for all x and equals $f(x)$ for all $x \neq 1$. Here, we have

$$\begin{aligned} \lim_{x \to 1} g(x) &= \lim_{x \to 1} \frac{x^2 + 2x - 3}{x - 1} \\ &= \lim_{x \to 1} (x + 3) = 4. \end{aligned}$$

Observe that if we choose $a = 4$, we now have that

$$\lim_{x \to 1} g(x) = 4 = g(1)$$

and so, g is continuous at $x = 1$.

Note that the graph of g is the same as the graph of f seen in Figure 2.23, except that we now include the point (1, 4). (See Figure 2.24.) Also, note that there's a very simple way to write $g(x)$. (Think about this.)

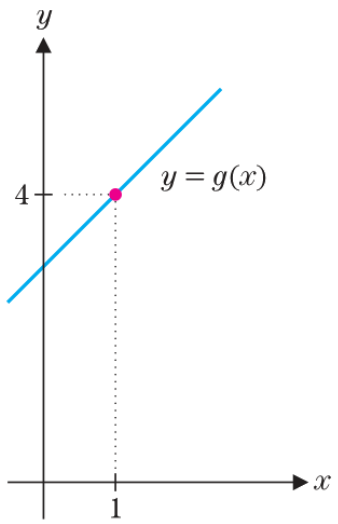


FIGURE 2.24
$y = g(x)$

◆ LESSON 2-5 · Limits Involving Infinity; Asymptotes

In exercises 5–22, determine each limit (answer as appropriate, with a number, ∞, −∞ or does not exist).

9. $\lim_{x\to\infty} \dfrac{x^2+3x-2}{3x^2+4x-1}$

10. $\lim_{x\to\infty} \dfrac{2x^2-x+1}{4x^2-3x-1}$

11. $\lim_{x\to-\infty} \dfrac{-x}{\sqrt{4+x^2}}$

12. $\lim_{x\to\infty} \dfrac{2x^2-1}{4x^3-5x-1}$

13. $\lim_{x\to\infty} \ln\left(\dfrac{x^2+1}{x-3}\right)$

14. $\lim_{x\to0^+} \ln(x\sin x)$

◆ LESSON 3-1 · Tangent Lines and Velocity

In exercises 19–22, the function represents the position in feet of an object at time t seconds. Find the average velocity between (a) $t = 0$ and $t = 2$, (b) $t = 1$ and $t = 2$, (c) $t = 1.9$ and $t = 2$, (d) $t = 1.99$ and $t = 2$, and (e) estimate the instantaneous velocity at $t = 2$.

19. $s(t) = 16t^2 + 10$

20. $s(t) = 3t^3 + t$

21. $s(t) = \sqrt{t^2 + 8t}$

22. $s(t) = 3\sin(t - 2)$

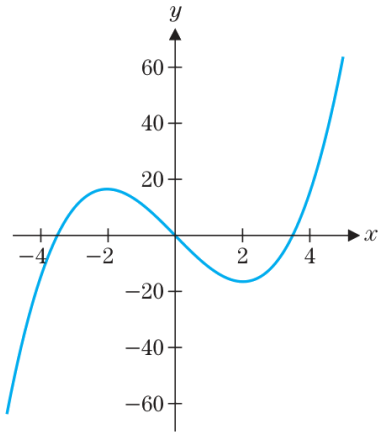


FIGURE 3.14
$y = f(x)$

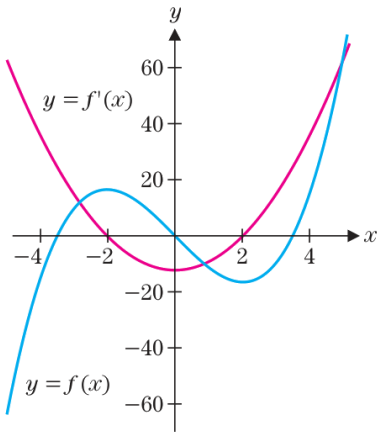


FIGURE 3.15
$y = f(x)$ and $y = f'(x)$

EXAMPLE 2.5 Sketching the Graph of f' Given the Graph of f

Given the graph of f in Figure 3.14, sketch a plausible graph of f'.

Solution Rather than worrying about exact values of $f'(x)$, we only wish to find the general shape of its graph. As in Figures 3.13a–3.13d, pick a few important points to analyze carefully. You should focus on any discontinuities and any places where the graph of f turns around.

The graph of $y = f(x)$ levels out at approximately $x = -2$ and $x = 2$. At these points, the derivative is 0. As we move from left to right, the graph rises for $x < -2$, drops for $-2 < x < 2$ and rises again for $x > 2$. This means that $f'(x) > 0$ for $x < -2$, $f'(x) < 0$ for $-2 < x < 2$ and finally $f'(x) > 0$ for $x > 2$. We can say even more. As x approaches -2 from the left, observe that the tangent lines get less steep. Therefore, $f'(x)$ becomes less positive as x approaches -2 from the left. Moving to the right from $x = -2$, the graph gets steeper until about $x = 0$, then gets less steep until it levels out at $x = 2$. Thus, $f'(x)$ gets more negative until $x = 0$, then less negative until $x = 2$. Finally, the graph gets steeper as we move to the right from $x = 2$. Putting this all together, we have the possible graph of f' shown in red in Figure 3.15, superimposed on the graph of f. ■

29. Graph and identify all x-values at which f is not differentiable. (a) $f(x) = |x| + |x - 2|$, (b) $f(x) = |x^2 - 4x|$

30. Graph and identify all x-values at which g is not differentiable. (a) $g(x) = e^{-2/x}$, (b) $g(x) = e^{-2/(x^3 - x)}$

THEOREM 3.2 (General Power Rule)

For any real number $r \neq 0$,

$$\frac{d}{dx}x^r = rx^{r-1}. \tag{3.7}$$

◆ LESSON 3-4 · The Product and Quotient Rules

In exercises 21–24, assume that f and g are differentiable with $f(0) = -1$, $f(1) = -2$, $f'(0) = -1$, $f'(1) = 3$, $g(0) = 3$, $g(1) = 1$, $g'(0) = -1$ and $g'(1) = -2$. Find an equation of the tangent line to the graph of $y = h(x)$ at $x = a$.

21. $h(x) = f(x)g(x)$;

(a) $a = 0$;

(b) $a = 1$

22. $h(x) = \dfrac{f(x)}{g(x)}$;

(a) $a = 1$;

(b) $a = 0$

23. $h(x) = x^2 f(x)$;

(a) $a = 1$;

(b) $a = 0$

24. $h(x) = \dfrac{x^2}{g(x)}$;

(a) $a = 1$;

(b) $a = 0$

◆ LESSON 3-4 · The Product and Quotient Rules

In exercises 21–24, assume that f and g are differentiable with $f(0) = -1$, $f(1) = -2$, $f'(0) = -1$, $f'(1) = 3$, $g(0) = 3$, $g(1) = 1$, $g'(0) = -1$ and $g'(1) = -2$. Find an equation of the tangent line to the graph of $y = h(x)$ at $x = a$.

21. $h(x) = f(x)g(x)$;

(a) $a = 0$;

(b) $a = 1$

22. $h(x) = \dfrac{f(x)}{g(x)}$;

(a) $a = 1$;

(b) $a = 0$

23. $h(x) = x^2 f(x)$;

(a) $a = 1$;

(b) $a = 0$

24. $h(x) = \dfrac{x^2}{g(x)}$;

(a) $a = 1$;

(b) $a = 0$

◆ LESSON 3-5 · The Chain Rule

In exercises 31 and 32, use the relevant information to compute the derivative for $h(x) = f(g(x))$.

31. $h'(1)$, where $f(1) = 3, g(1) = 2, f'(1) = 4, f'(2) = 3, g'(1) = -2$ and $g'(3) = 5$

32. $h'(2)$, where $f(2) = 1, g(2) = 3, f'(2) = -1, f'(3) = -3, g'(1) = 2$ and $g'(2) = 4$

33. A function f is an **even function** if $f(-x) = f(x)$ for all x and is an **odd function** if $f(-x) = -f(x)$ for all x. Prove that the derivative of an even function is odd and the derivative of an odd function is even.

34. If the graph of a differentiable function f is symmetric about the line $x = a$, what can you say about the symmetry of the graph of f'?

◆ LESSON 3-5 · The Chain Rule

In exercises 35–38, find the derivative where f is an unspecified differentiable function.

35. (a) $f(x^2)$

(b) $[f(x)]^2$

(c) $f(f(x))$

36. (a) $f(\sqrt{x})$

(b) $\sqrt{f(x)}$

(c) $f(xf(x))$

◆ LESSON 3-5 · The Chain Rule

In exercises 17–22, f has an inverse g. Use Theorem 5.2 to find $g'(a)$.

17. $f(x) = x^3 + 4x - 1, a = -1$

18. $f(x) = x^5 + 4x - 2, a = -2$

19. $f(x) = x^5 + 3x^3 + x, a = 5$

20. $f(x) = x^3 + 2x + 1, a = -2$

21. $f(x) = \sqrt{x^3 + 2x + 4}, a = 2$

22. $f(x) = \sqrt{x^5 + 4x^3 + 3x + 1}, a = 3$

◆ LESSON 3-6 · Derivatives of Trigonometric Functions

In exercises 1–18, find the derivative of each function.

9. $f(t) = \sin 3t \sec 3t$

10. $f(t) = \sqrt{\cos 5t \sec 5t}$

11. $f(w) = \dfrac{1}{\sin 4w}$

12. $f(w) = w^2 \sec^2 3w$

◆ LESSON 3-6 · Derivatives of Trigonometric Functions

13. $f(x) = 2 \sin 2x \cos 2x$

14. $f(x) = 4 \sin^2 3x + 4 \cos^2 3x$

15. $f(x) = \tan \sqrt{x^2 + 1}$

16. $f(x) = 4x^2 \sin x \sec 3x$

17. $f(x) = \sin^3 \left(\cos \sqrt{x^3 + 2x^2} \right)$

18. $f(x) = \tan^4(\sin^2(x^3 + 2x))$

◆ LESSON 3-7 · Derivatives of Exponential and Logarithmic Functions

In exercises 29 and 30, find all values of x for which the tangent line to $y = f(x)$ is horizontal.

29. (a) $f(x) = xe^{-2x}$

(b) $f(x) = xe^{-3x}$

30. (a) $f(x) = x^2e^{-2x}$

(b) $f(x) = x^2e^{-3x}$

In exercises 1–24, differentiate each function.

20. (a) $f(x) = \dfrac{\sqrt{\ln x}}{x}$

(b) $g(t) = \dfrac{\ln \sqrt{t}}{t}$

21. (a) $h(x) = e^x \ln x$

(b) $f(x) = e^{\ln x}$

22. (a) $h(x) = 2^{e^x}$

(b) $f(x) = \dfrac{e^x}{2^x}$

EXAMPLE 8.2 Finding a Tangent Line by Implicit Differentiation

Find $y'(x)$ for $x^2y^2 - 2x = 4 - 4y$. Then, find an equation of the tangent line at the point $(2, -2)$.

Solution Differentiating both sides with respect to x, we get

$$\frac{d}{dx}(x^2y^2 - 2x) = \frac{d}{dx}(4 - 4y).$$

Since the first term is the product of x^2 and y^2, we must use the product rule. We get

$$2xy^2 + x^2(2y)y'(x) - 2 = 0 - 4y'(x).$$

Grouping the terms with $y'(x)$ on one side, we get

$$(2x^2y + 4)y'(x) = 2 - 2xy^2,$$

so that

$$y'(x) = \frac{2 - 2xy^2}{2x^2y + 4}.$$

Substituting $x = 2$ and $y = -2$, we get the slope of the tangent line,

$$y'(2) = \frac{2 - 16}{-16 + 4} = \frac{7}{6}.$$

Finally, an equation of the tangent line is given by

$$y + 2 = \frac{7}{6}(x - 2).$$

We have plotted the curve and the tangent line at $(2, -2)$ in Figure 3.41 using the implicit plot mode of our computer algebra system.

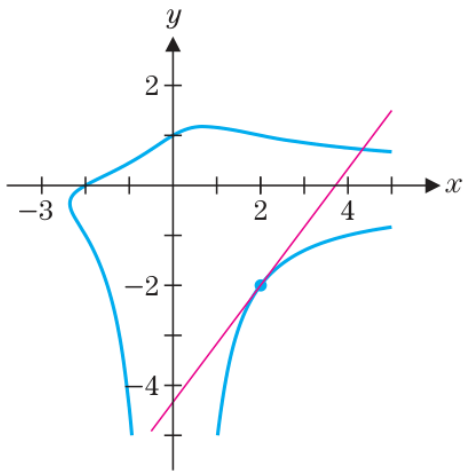


FIGURE 3.41
Tangent line at $(2, -2)$

In exercises 29–34, find the derivative of the given function.

29. (a) $f(x) = \sin^{-1}(x^3 + 1)$

(b) $f(x) = \sin^{-1}(\sqrt{x})$

30. (a) $f(x) = \cos^{-1}(x^2 + x)$

(b) $f(x) = \cos^{-1}(2/x)$

31. (a) $f(x) = \tan^{-1}(\sqrt{x})$

(b) $f(x) = \tan^{-1}(1/x)$

32. (a) $f(x) = \sqrt{2 + \tan^{-1} x}$

(b) $f(x) = e^{\tan^{-1} x}$

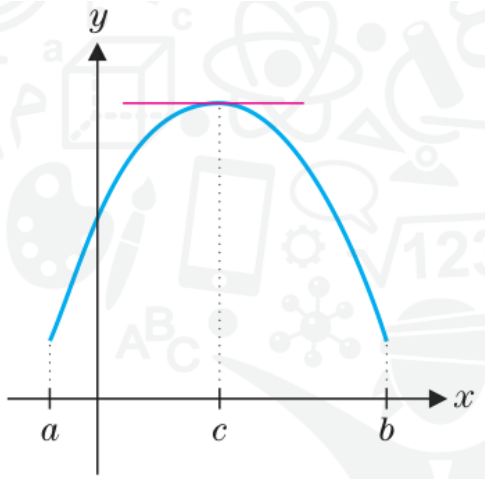


FIGURE 3.48a
Graph initially rising

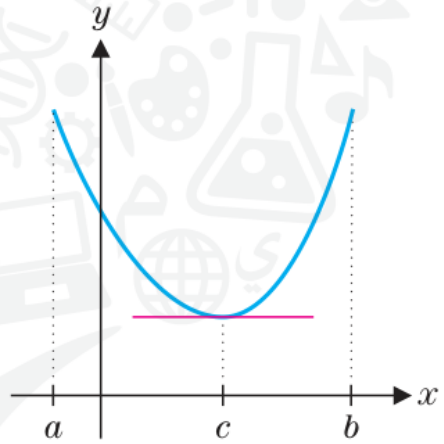


FIGURE 3.48b
Graph initially falling

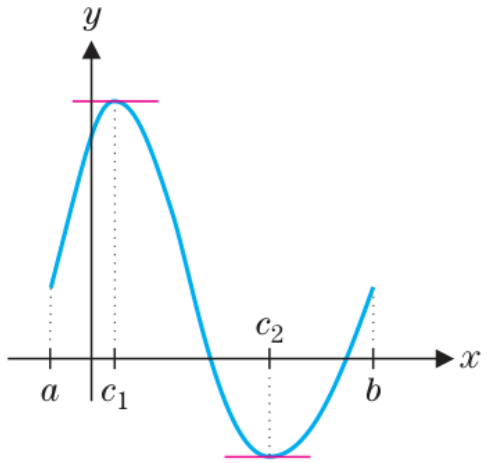


FIGURE 3.48c
Graph with two horizontal tangents

Note that since $f'(x) = 0$ at a horizontal tangent, this says that there is at least one point c in (a, b), for which $f'(c) = 0$. (See Figures 3.48a to 3.48c.)

THEOREM 10.1 (Rolle's Theorem)

Suppose that f is continuous on the interval $[a, b]$, differentiable on the interval (a, b) and $f(a) = f(b)$. Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

This suggests the following definition of continuity at a point.

DEFINITION 4.1

For a function f defined on an open interval containing $x = a$, we say that f is **continuous** at a when

$$\lim_{x \to a} f(x) = f(a).$$

Otherwise, f is said to be **discontinuous** at $x = a$.

29. Suppose that a state's income tax code states that the tax liability on x dirhams of taxable income is given by

$$T(x) = \begin{cases} 0 & \text{if } x \le 0 \\ 0.14x & \text{if } 0 < x < 10{,}000 \\ c + 0.21x & \text{if } 10{,}000 \le x. \end{cases}$$

Determine the constant c that makes this function continuous for all x. Give a rationale why such a function should be continuous.

30. Suppose a state's income tax code states that tax liability is 12% on the first AED 20,000 of taxable earnings and 16% on the remainder. Find constants a and b for the tax function

$$T(x) = \begin{cases} 0 & \text{if } x \le 0 \\ a + 0.12x & \text{if } 0 < x \le 20{,}000 \\ b + 0.16(x - 20{,}000) & \text{if } x > 20{,}000 \end{cases}$$

such that $T(x)$ is continuous for all x.

31. In example 4.8, find b and c to complete the table.

32. In example 4.8, show that $T(x)$ is continuous for $x = 6000$.

◆ LESSON 2-5 · Limits Involving Infinity; Asymptotes

In exercises 29–32, determine all vertical and slant asymptotes.

29. $y = \dfrac{x^3}{4 - x^2}$

30. $y = \dfrac{x^2 + 1}{x - 2}$

31. $y = \dfrac{x^3}{x^2 + x - 4}$

32. $y = \dfrac{x^4}{x^3 + 2}$

◆ LESSON 3-4 · The Product and Quotient Rules

In exercises 17–20, find an equation of the tangent line to the graph of $y = f(x)$ at $x = a$.

17. $f(x) = (x^2 + 2x)(x^4 + x^2 + 1), a = 0$

18. $f(x) = (x^3 + x + 1)(3x^2 + 2x - 1), a = 1$

19. $f(x) = \dfrac{x+1}{x+2}, a = 0$

20. $f(x) = \dfrac{x+3}{x^2+1}, a = 1$

◆ LESSON 3-6 · Derivatives of Trigonometric Functions

In exercises 1–18, find the derivative of each function.

3. $f(t) = \tan^3 2t - \csc^4 3t$

4. $f(t) = t^2 + 2\cos^2 4t$

5. $f(x) = x\cos 5x^2$

6. $f(x) = x^2 \sec 4x$

7. $f(x) = \dfrac{\sin x^2}{x^2}$

8. $f(x) = \dfrac{x^2}{\csc^4 2x}$

◆ LESSON 3-7 · Derivatives of Exponential and Logarithmic Functions

35. A bacterial population starts at 200 and triples every day. Find a formula for the population after t days and find the percentage rate of change in population.

36. A bacterial population starts at 500 and doubles every four days. Find a formula for the population after t days and find the percentage rate of change in population.

37. The concentration of a certain chemical after t seconds of an autocatalytic reaction is given by $c(t) = \dfrac{6}{2e^{-8t} + 1}$. Show that $c'(t) > 0$ and use this information to determine that the concentration of the chemical never exceeds 6.

38. The concentration of a certain chemical after t seconds of an autocatalytic reaction is given by $c(t) = \dfrac{10}{9e^{-10t} + 2}$. Show that $c'(t) > 0$ and use this information to determine that the concentration of the chemical never exceeds 5.

◆ LESSON 3-10 · The Mean Value Theorem

In exercises 1–6, check the hypotheses of Rolle's Theorem and the Mean Value Theorem and find a value of c that makes the appropriate conclusion true. Illustrate the conclusion with a graph.

1. $f(x) = x^2 + 1, [-2, 2]$

2. $f(x) = x^2 + 1, [0, 2]$

3. $f(x) = x^3 + x^2, [0, 1]$

4. $f(x) = x^3 + x^2, [-1, 1]$

5. $f(x) = \sin x, [0, \pi/2]$

6. $f(x) = \sin x, [-\pi, 0]$